Marcus Hurt
Maria Vilanova Garcia
Kenneth Sutter
Aditi Thangavelu
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Group 2 Project Proposal: Repository-aware code completion

We will be investigating how code completion works within the greater context of the entire repository. Our team would like to see how information from other files affects a model's ability to complete code. This is interesting for a few reasons. First is that research in this field has exploded over the last couple of years making it a hot topic in the AI space. Secondly real software in working environments depends on functions, classes, and various coding structures spread out across many files.

Our group will review research in the paper CodeRL: Mastering Code Generation through retrained Models and Deep Reinforcement Learning by Hung Le, Yue Wang, Akhilesh Deepak Gotmare, Silvio Savarese, and Steven C.H. Hoi. We will also look into CrossCodeEval: A Diverse and Multilingual Benchmark for Cross-File Code Completion by Yangruibo Ding, Zijian Wang, Wasi Uddin Ahmad, et al.

Currently we plan to use existing data from CrossCodeEval which includes code completion examples from real repositories that require information from other files. We also have the option of using RepoBench which is Python and Java data for evaluating cross file retrieval and repository level code completion. We have not decided yet on exactly which one.

The team has not decided whether to use a small pretrained code model without additional training or fine tune one ourselves using LoRA. If we decide to use a model without training our modifications will compare short snips of code from multiple files against large snips of code from fewer files under the same context budget. If we go the fine tuning route we will train separate Python examples containing cross file context and compare the fine tuned model with the original model.

We will evaluate our results in two ways qualitatively and quantitatively. The team will measure exact match accuracy, edit similarity, and completion time. These should be able to show us how close the output is matching the reference code and how long the generation time is.